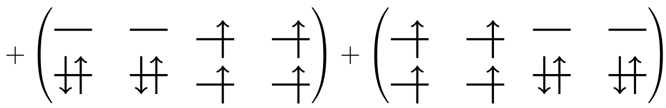
1 Linear Combinations

$$S * Q + Q * S$$



written in CI vectors:

$$+1 \cdot |0220| \cdot |aaaa| + 1 \cdot |aaaa| \cdot |0220|$$

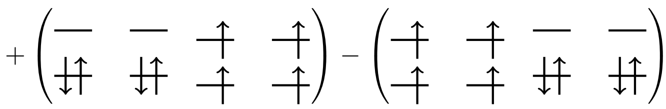
monomer determinants:

$$+ \quad 1 \quad \cdot \quad \left|\left(b^r_{1u}\right)\left(b^r_{2g}\right)\left(b^r_{3g}\right)\left(a^r_u\right)\right| \quad + \quad 1 \quad \cdot \quad \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^l_{3g}\right)\left(a^l_u\right)\right|$$

multiplied out:

$$+1 \cdot |(-a_g + b_{1u})(-b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(+b_{2u} + b_{3g})(+b_{1g} + a_u)|$$

$$S * Q - Q * S$$



written in CI vectors:

$$+1 \cdot |0220| \cdot |aaaa| - 1 \cdot |aaaa| \cdot |0220|$$

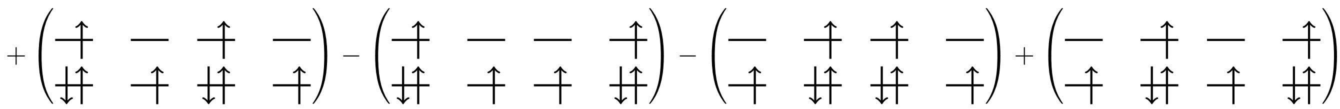
monomer determinants:

$$+ \quad 1 \quad \cdot \quad \left|\left(b^r_{1u}\right)\left(b^r_{2g}\right)\left(b^r_{3g}\right)\left(a^r_u\right)\right| \quad - \quad 1 \quad \cdot \quad \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^l_{3g}\right)\left(a^l_u\right)\right|$$

multiplied out:

$$+1 \cdot |(-a_g + b_{1u})(-b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(+b_{2u} + b_{3g})(+b_{1g} + a_u)|$$

$$i^3 b_{2u} * i^3 b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |0a2a| \cdot |0a2a|$$

monomer determinants:

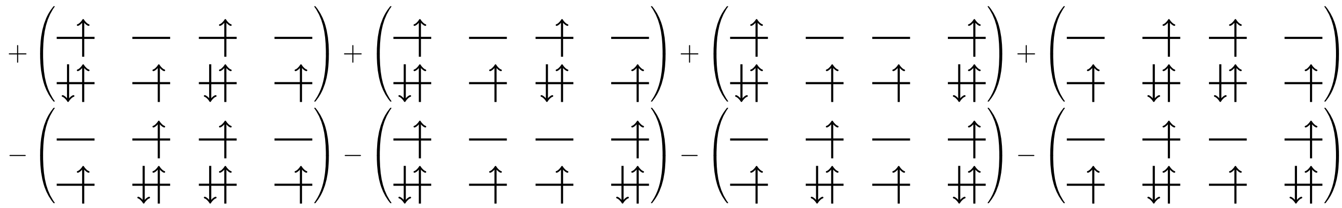
$$+ \ 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{2g}\right)\left(a^r_u\right)\right| - 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_u\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_u\right)\left(b^r_{2g}\right)\left(a^r_u\right)\right|$$

multiplied out:

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| - 1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$i^3b_{2u} * e^3b_{2u} + e^3b_{2u} * i^3b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |a2a0|$$

$$-1 \cdot |0a2a| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{2g}\right)\left(a\right.\right.$$

$$-1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

multiplied out:

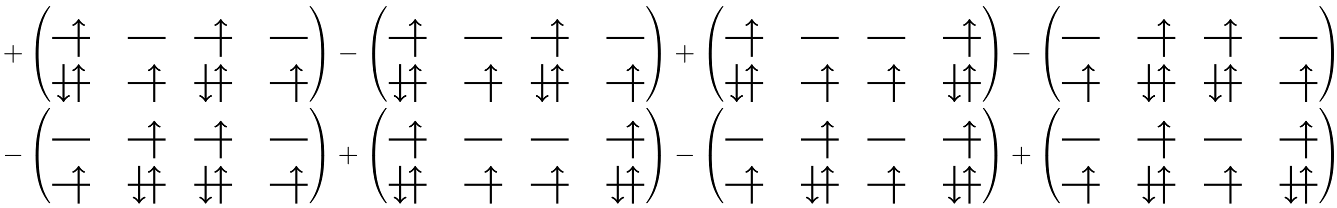
$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| - 1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$i^3b_{2u} * e^3b_{2u} - e^3b_{2u} * i^3b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |a2a0|$$

$$-1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{2g}\right)\left(a\right.\right.$$

$$-1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

multiplied out:

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| - 1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |02aa| \cdot |a2a0|$$

$$-1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |02aa| + 1 \cdot |02aa| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \big|\left(b^l_{2g}\right)\big(a$$

$$+1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

multiplied out:

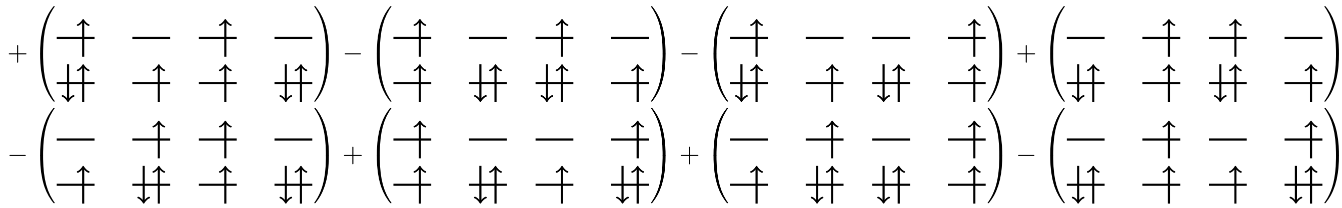
$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$i^3b_{2u} * e^3b_{3u} - e^3b_{3u} * i^3b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |02aa| \cdot |a2a0|$$

$$-1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |02aa| - 1 \cdot |02aa| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{2g}\right)\left(a\right.\right.$$

$$+1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

multiplied out:

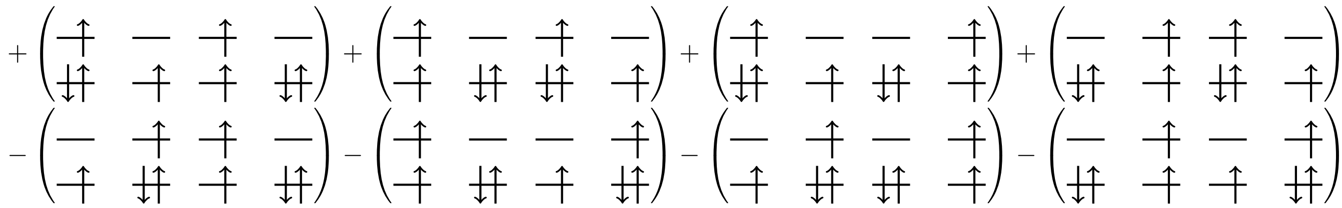
$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$i^3b_{2u} * i^3b_{3u} + i^3b_{3u} * i^3b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |02aa| \cdot |a2a0|$$

$$-1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |02aa| - 1 \cdot |02aa| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right| + 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| + 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{3g}^r\right)\left(a_{u}^r\right)\right| + 1 \cdot \left|\left(b_{3g}^l\right)\left(a_{u}^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| - 1 \cdot \left|\left(b_{2g}^l\right)\left(a\right.\right.$$

$$-1 \cdot \left|\left(b_{2g}^l\right)\left(a_{u}^l\right)\left(b_{3g}^r\right)\left(a_{u}^r\right)\right| - 1 \cdot \left|\left(b_{3g}^l\right)\left(a_{u}^l\right)\left(b_{2g}^r\right)\left(a_{u}^r\right)\right|$$

multiplied out:

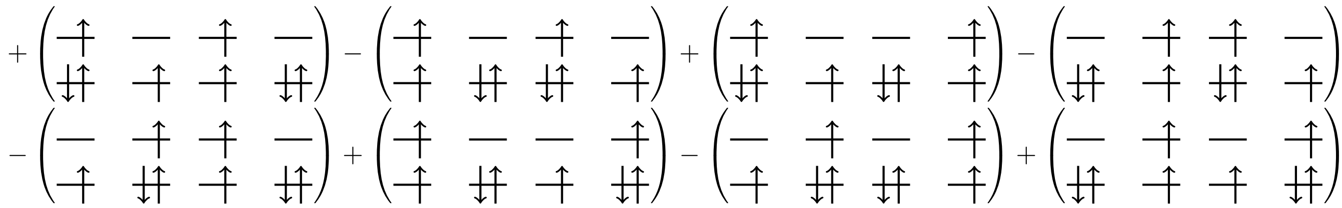
$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$i^3b_{2u} * i^3b_{3u} - i^3b_{3u} * i^3b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |02aa| \cdot |a2a0|$$

$$-1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |02aa| + 1 \cdot |02aa| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right| - 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| + 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| - 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| - 1 \cdot \left|\left(b_{2g}^l\right)\left(a\right.\right.$$

$$-1 \cdot \left|\left(b_{2g}^l\right)\left(a_u^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| + 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{2g}^r\right)\left(a_u^r\right)\right|$$

multiplied out:

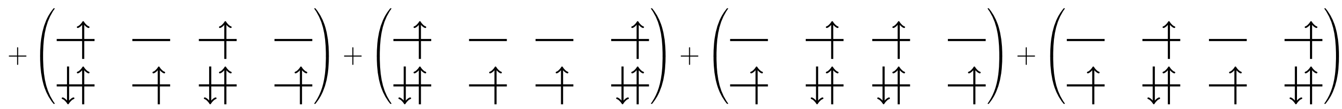
$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$e^3b_{2u} * e^3b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |0a2a| \cdot |0a2a|$$

monomer determinants:

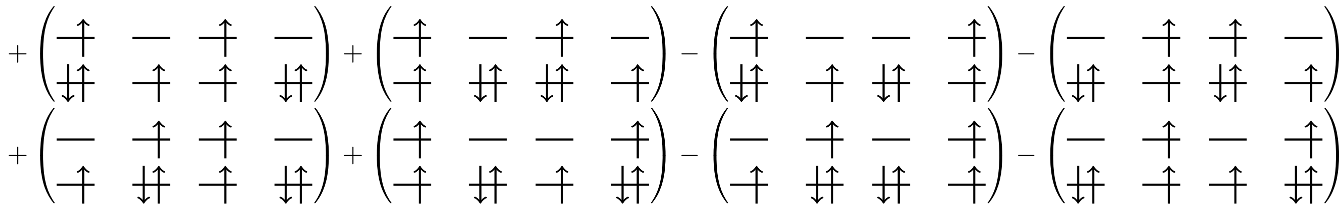
$$+\ 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| + 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{2g}^r\right)\left(a_u^r\right)\right| + 1 \cdot \left|\left(b_{2g}^l\right)\left(a_u^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| + 1 \cdot \left|\left(b_{2g}^l\right)\left(a_u^l\right)\left(b_{2g}^r\right)\left(a_u^r\right)\right|$$

multiplied out:

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$e^3b_{2u} * e^3b_{3u} + e^3b_{3u} * e^3b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |02aa| \cdot |a2a0|$$

$$+1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |02aa| - 1 \cdot |02aa| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a\right.\right.$$

$$-1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

multiplied out:

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$e^3b_{2u} * e^3b_{3u} - e^3b_{3u} * e^3b_{2u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$+\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |02aa| \cdot |a2a0|$$

$$+1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |02aa| + 1 \cdot |02aa| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right| - 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| - 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| + 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| + 1 \cdot \left|\left(b_{2g}^l\right)\left(a\right.\right.$$

$$-1 \cdot \left|\left(b_{2g}^l\right)\left(a_u^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| + 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{2g}^r\right)\left(a_u^r\right)\right|$$

multiplied out:

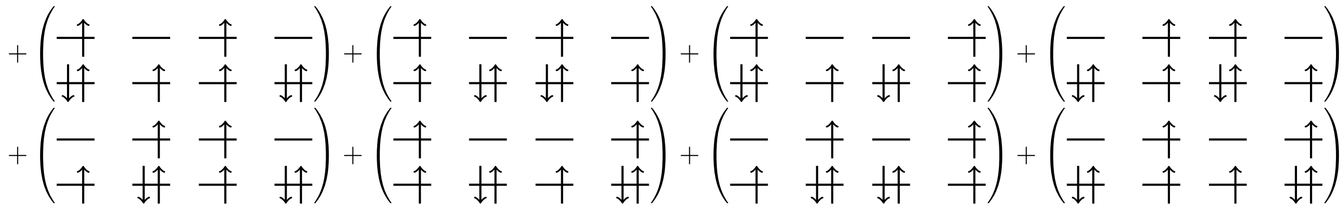
$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$e^3b_{2u} * i^3b_{3u} + i^3b_{3u} * e^3b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |02aa| \cdot |a2a0|$$

$$+1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |02aa| + 1 \cdot |02aa| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a\right.\right.$$

$$+1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

multiplied out:

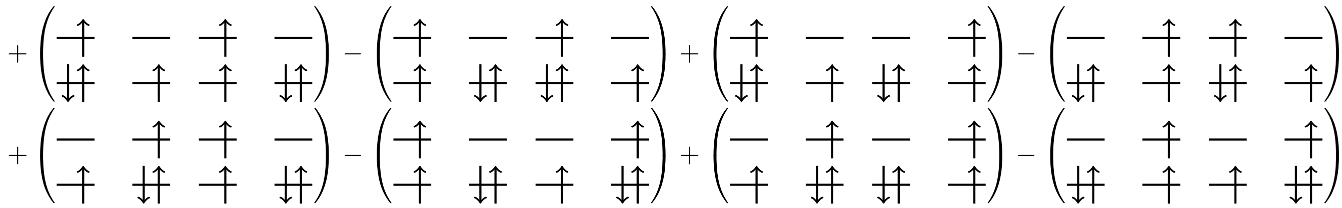
$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$e^3b_{2u} * i^3b_{3u} - i^3b_{3u} * e^3b_{2u}$$



written in CI vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |02aa| \cdot |a2a0|$$

$$+1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |02aa| - 1 \cdot |02aa| \cdot |0a2a|$$

monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right| - 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| + 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{3g}^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| - 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{1u}^r\right)\left(b_{3g}^r\right)\right| + 1 \cdot \left|\left(b_{2g}^l\right)\left(a\right.\right.$$

$$+1 \cdot \left|\left(b_{2g}^l\right)\left(a_u^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| - 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{2g}^r\right)\left(a_u^r\right)\right|$$

multiplied out:

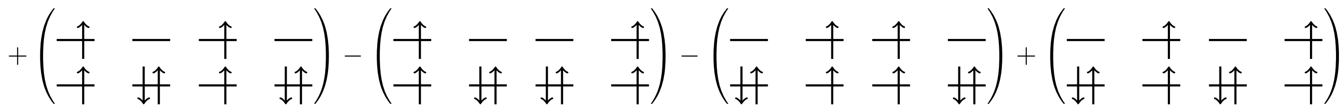
$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$e^3b_{3u} * e^3b_{3u}$$



written in CI vectors:

$$+1 \cdot |aa20| \cdot |aa20| - 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |aa20| + 1 \cdot |02aa| \cdot |02aa|$$

monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right| - 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| - 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right| + 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right|$$

multiplied out:

$$+1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

$$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:

$$+1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| + 1 \cdot |02aa| \cdot |aa20|$$

$$-1 \cdot |02aa| \cdot |aa20| - 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa|$$

monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| - 1 \cdot \left|\left(b^l_{3g}\right)\left(a\right.\right.$$

$$-1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right|$$

multiplied out:

$$+1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})|$$

$$-1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

$$e^3b_{3u} * i^3b_{3u} - i^3b_{3u} * e^3b_{3u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$
$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:

$$+1 \cdot |aa20| \cdot |aa20| - 1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |aa20|$$

$$-1 \cdot |02aa| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa| + 1 \cdot |02aa| \cdot |02aa|$$

monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right| - 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right| + 1 \cdot \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| - 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right| - 1 \cdot \left|\left(b_{3g}^l\right)\left(a\right.\right.$$

$$-1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| + 1 \cdot \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right|$$

multiplied out:

$$+1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})|$$

$$-1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

$$i^3b_{3u} * i^3b_{3u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:

$$+1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| + 1 \cdot |02aa| \cdot |aa20| + 1 \cdot |02aa| \cdot |02aa|$$

monomer determinants:

$$+\ 1\ \cdot\ \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right|\ +\ 1\ \cdot\ \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right|\ +\ 1\ \cdot\ \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{1u}^r\right)\left(b_{2g}^r\right)\right|\ +\ 1\ \cdot\ \left|\left(b_{3g}^l\right)\left(a_u^l\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right|$$

multiplied out:

$$+1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$
